Unit –III

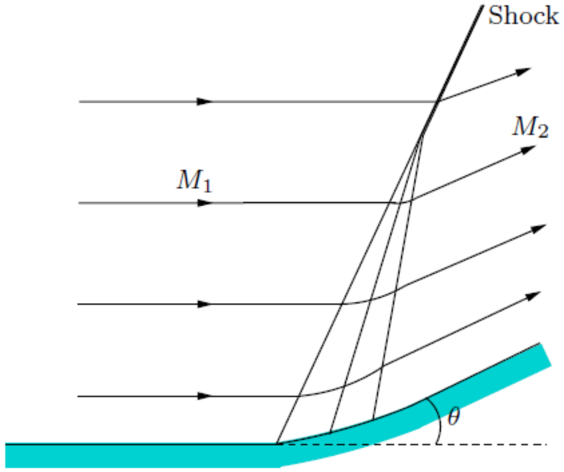
Expansion waves: Supersonic compression and supersonic expansion waves, Prandtl-Meyer Expansion Function (No Derivation), Shock expansion theory, Wave reflection and wave intersection shock system

Supersonic Compression

In general Non isentropic.

But compression process can be regarded isentropic, when the turning of the flow is achieved through large number of weak oblique shocks

It is termed as ***continuous compression***

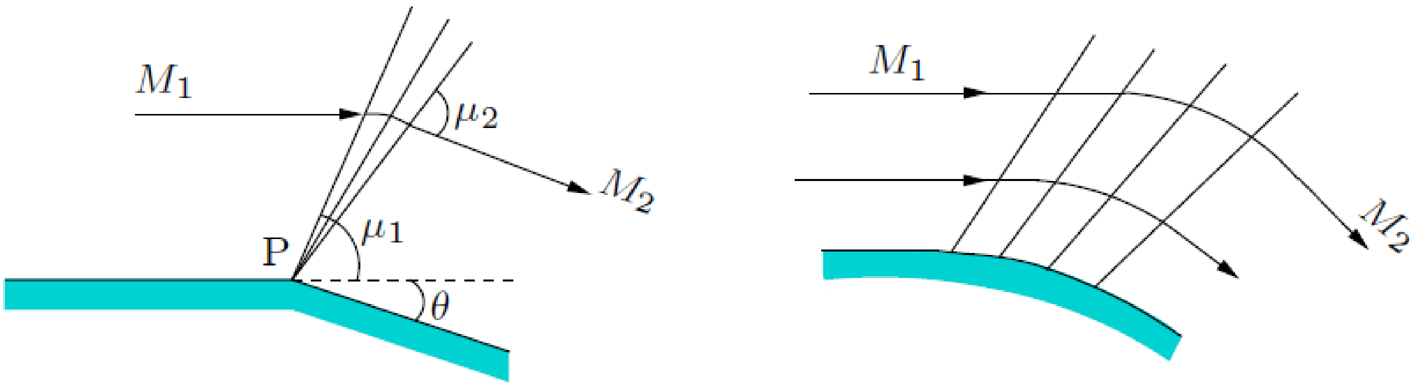


Smooth continuous compression.

The flow turning taking place through n weak compression waves, each wave turning the flow by an angle $\Delta\theta$

$$(s_n - s_1) \sim n(\Delta\theta)^3 \sim n\Delta\theta(\Delta\theta)^2 \sim \theta(\Delta\theta)^2$$

Supersonic Expansion by Turning



(a) Centred expansion (b) Continuous (simple) expansion

Expansion rays in an expansion fan are isentropic waves across which the change of pressure, temperature, density and Mach number are small but finite

But at Vertex P

Entropy change associated with the expansion process at point p is finite. Hence at P its non isentropic

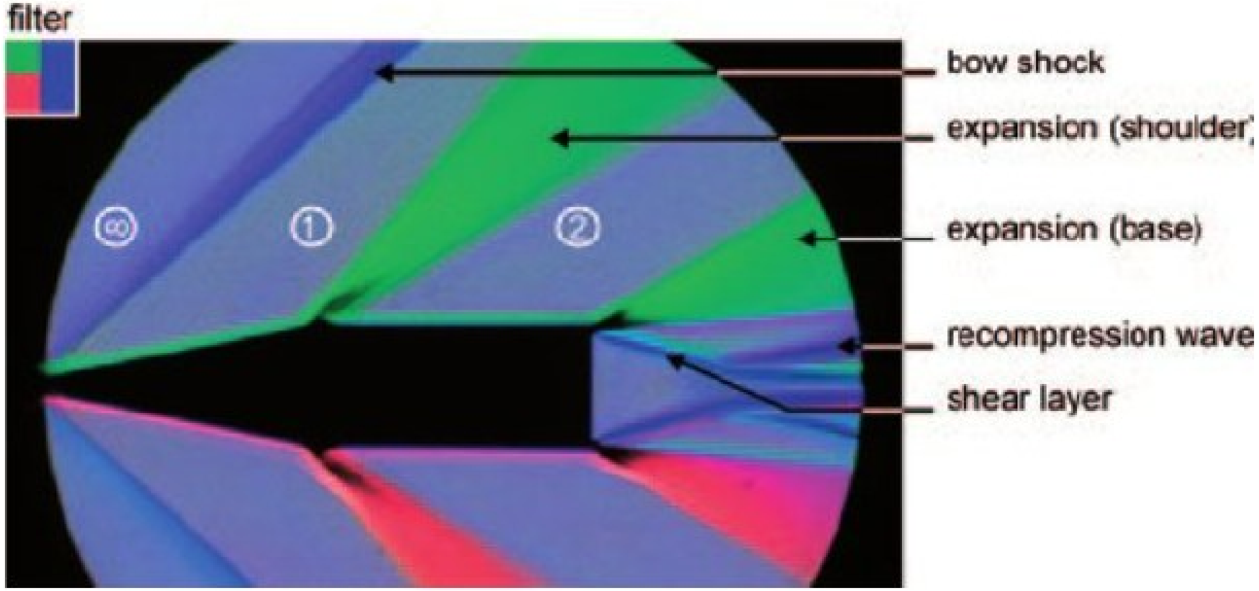
Centered expansion process is isentropic everywhere except at the vertex

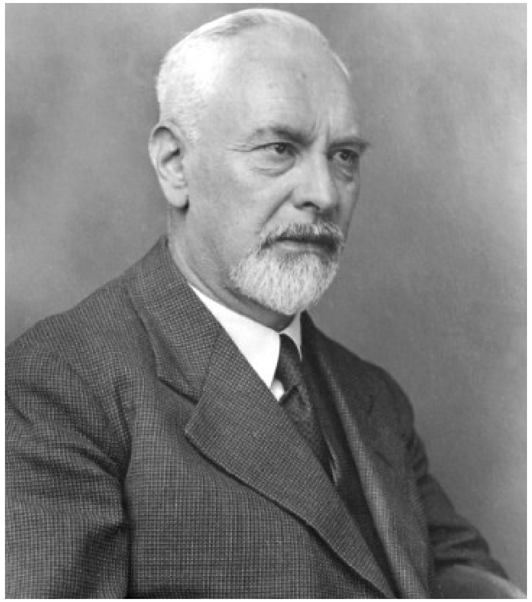
Continuous expansion

No sudden change of flow properties

The continuous expansion is isentropic everywhere

The Centered wave, also called a Prandtl-Meyer expansion fan

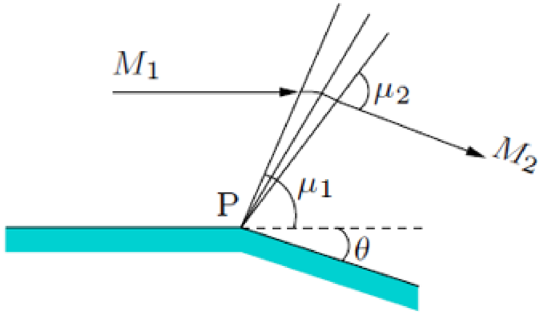




Ludwig Prandtl

Lothar Meyer

The Prandtl-Meyer Expansion



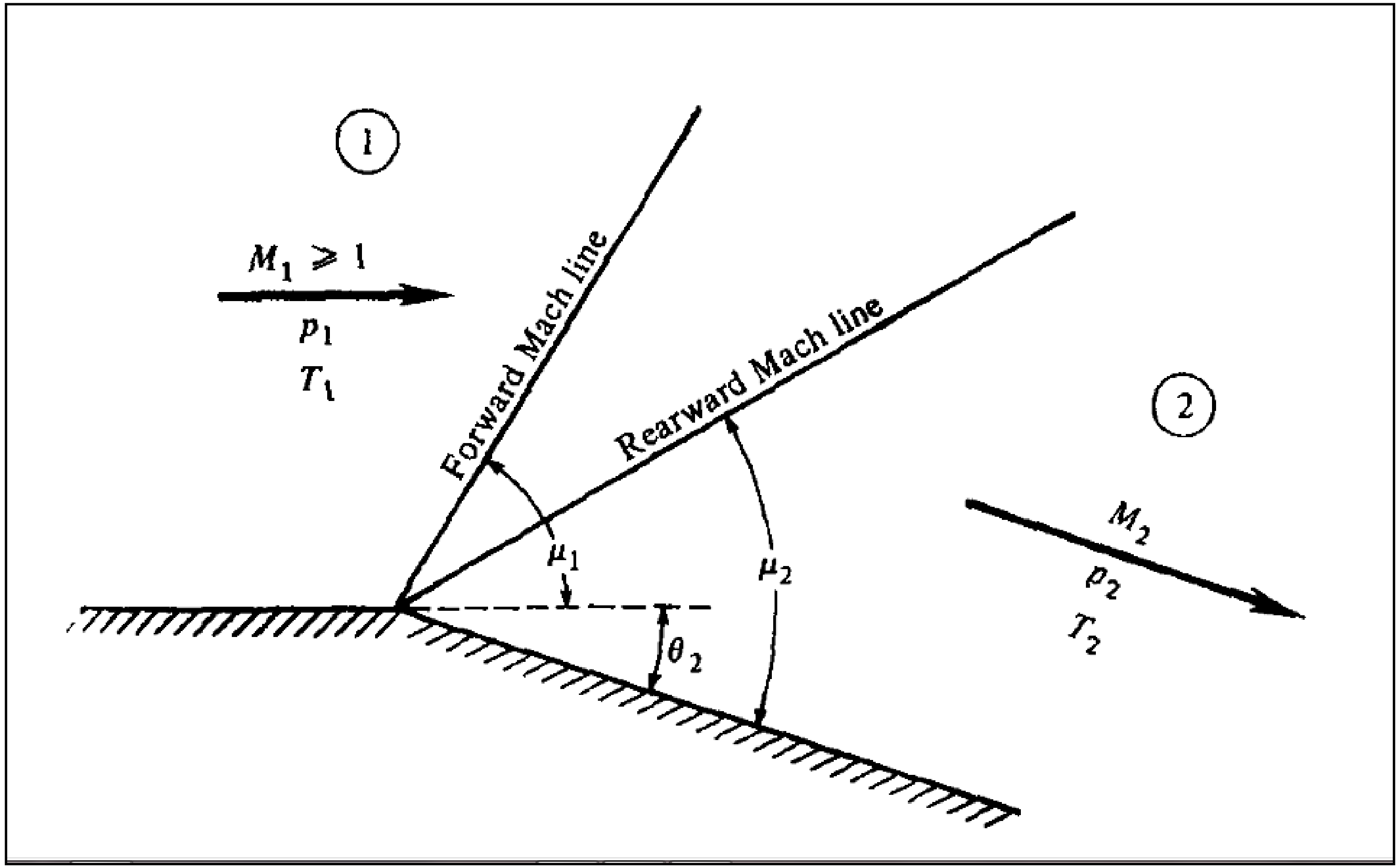
(a) Centred expansion

Flow passing through a Prandtl-Meyer expansion would experience a smooth, gradual change of flow properties

The Prandtl-Meyer fan consists of an infinite number of Mach (expansion) lines, centered at the convex corner

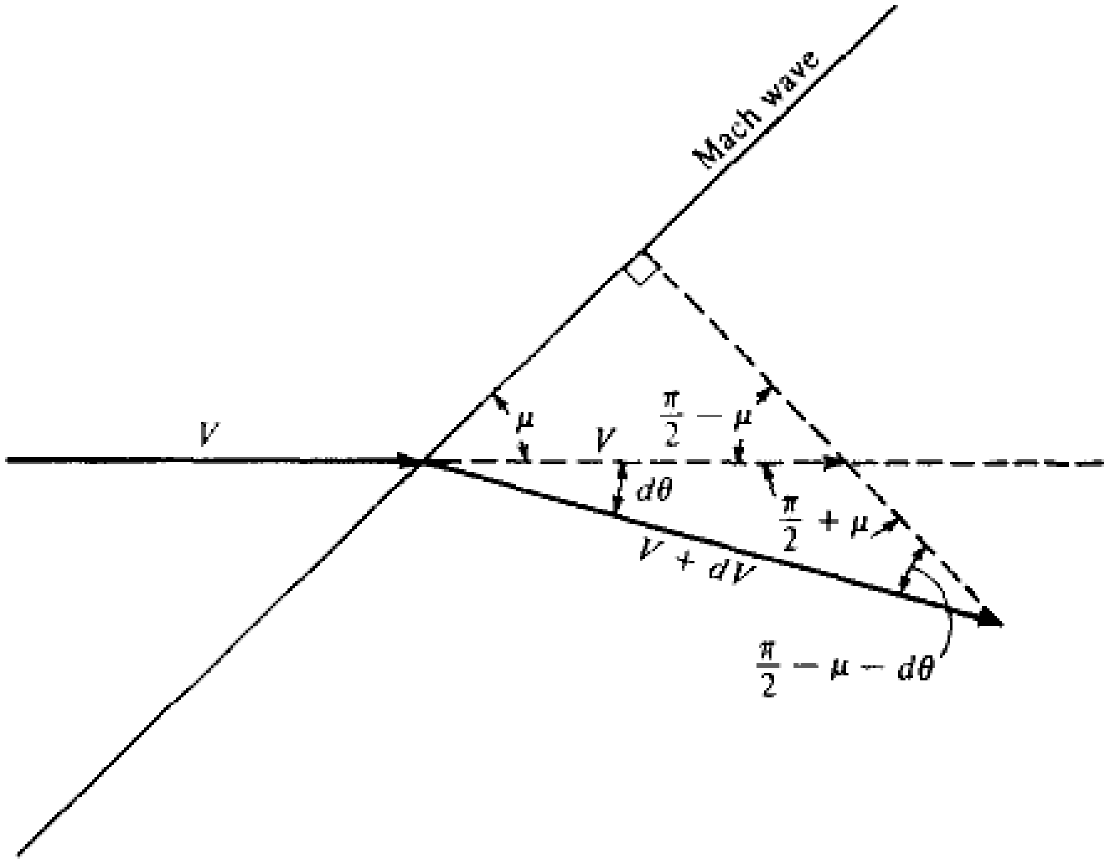
There are two Mach lines, one at an angle μ_1 to the initial flow (upstream of fan) direction and the other at an angle μ_2 to the final flow (downstream of fan) direction

Expansion wave



The Prandtl-Meyer Function

A flow which preserves its own geometry in space or time or both is called a *self-similar flow*. Such motions are described by a single independent variable, referred to as similarity variable. The Prandtl- Meyer function is such a *similarity variable*



$$\frac{V + dV}{V} = \frac{\sin(\pi/2 + \mu)}{\sin(\pi/2 - \mu - d\theta)}$$

$$\sin\left(\frac{\pi}{2} + \mu\right) = \sin\left(\frac{\pi}{2} - \mu\right) = \cos\mu$$

$$\sin\left(\frac{\pi}{2} - \mu - d\theta\right) = \cos(\mu + d\theta) = \cos\mu\cos d\theta - \sin\mu\sin d\theta$$

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos\mu\cos d\theta - \sin\mu\sin d\theta}$$

$$1 + \frac{dV}{V} = \frac{\cos\mu}{\cos\mu - d\theta\sin\mu} = \frac{1}{1 - d\theta\tan\mu}$$

series expansion (for $x < 1$),

$$\frac{1}{1 - x} = 1 + x + x^2 + x^3 + \cdots$$

$$1 + \frac{dV}{V} = 1 + d\theta \tan \mu + \cdots$$

$$d\theta = \frac{dV/V}{\tan \mu}$$

$$\mu = \sin^{-1} \frac{1}{M}$$

$$\tan \mu = \frac{1}{\sqrt{M^2 - 1}} \qquad \boxed{d\theta = \sqrt{M^2 - 1}\,\frac{dV}{V}}$$

$$\int_{\theta_1}^{\theta_2} d\theta = \int_{M_1}^{M_2} \sqrt{M^2 - 1}\, \frac{dV}{V}$$

$$V = Ma$$

$$\ln V = \ln M + \ln a$$

$$\frac{dV}{V} = \frac{dM}{M} + \frac{da}{a}$$

$$a = a_o \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1/2}$$

$$\frac{da}{a} = -\left(\frac{\gamma - 1}{2}\right) M \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1} dM$$

$$\frac{dV}{V} = \frac{1}{1 + \frac{\gamma - 1}{2} M^2} \frac{dM}{M}$$

$$\int_{\theta_1}^{\theta_2} d\theta = \theta_2 - 0 = \int_{M_1}^{M_2} \frac{\sqrt{M^2 - 1}}{1 + \frac{\gamma - 1}{2} M^2} \frac{dM}{M}$$

$$\nu(M) = \int \frac{\sqrt{M^2 - 1}}{1 + \frac{\gamma - 1}{2} M^2} \frac{dM}{M}$$

$$\boxed{\nu(M) = \sqrt{\frac{\gamma + 1}{\gamma - 1}} \tan^{-1} \sqrt{\frac{\gamma - 1}{\gamma + 1}(M^2 - 1)} - \tan^{-1} \sqrt{M^2 - 1}}$$

$$\boxed{\theta_2 = \nu(M_2) - \nu(M_1)}$$

M_1 varies from 1 to ∞

ν = 0

$$\nu_{\max} = \frac{\pi}{2}\left(\sqrt{\frac{\gamma + 1}{\gamma - 1}} - 1\right)$$

For air with $\gamma = 1.4$, $\nu_{\max} = 130.5^\circ$ for $M_1 = \infty$

Calculation procedure

1. Obtain $\nu(M_1)$ from Table .   for the given M_1.

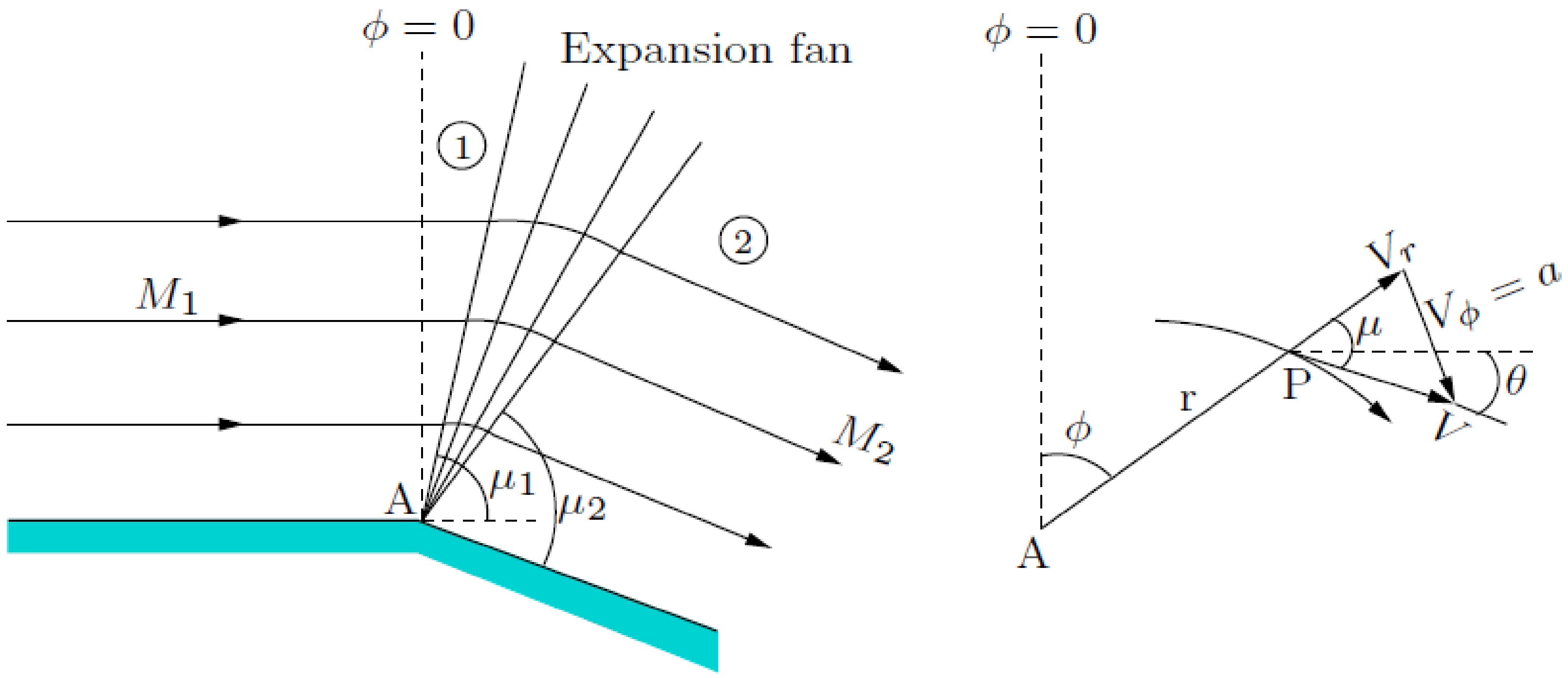
2. Calculate $\nu(M_2)$ using the given θ_2 and $\nu(M_1)$ obtained

3. Obtain M_2

$$\frac{T_1}{T_2} = \frac{1 + \frac{\gamma - 1}{2} M_2^2}{1 + \frac{\gamma - 1}{2} M_1^2}$$

$$\frac{p_1}{p_2} = \left[\frac{1 + \frac{\gamma - 1}{2} M_2^2}{1 + \frac{\gamma - 1}{2} M_1^2}\right]^{\gamma/(\gamma-1)}$$

Expansion fan Analysis in polar coordinate r and Φ



In polar coordinates r and ϕ

$$\frac{\partial}{\partial r}(\rho r V_r) + \frac{\partial}{\partial \phi}(\rho V_\phi) = 0$$

Euler's equation (Inviscid Momentum Eqn)

$$V_r \frac{\partial V_r}{\partial r} + \frac{V_\phi}{r}\frac{\partial V_r}{\partial \phi} - \frac{V_\phi^2}{r} = -\frac{1}{\rho}\frac{\partial p}{\partial r}$$

$$V_r \frac{\partial V_\phi}{\partial r} + \frac{V_\phi}{r}\frac{\partial V_\phi}{\partial \phi} + \frac{V_r V_\phi}{r} = -\frac{1}{\rho}\frac{1}{r}\frac{\partial p}{\partial \phi}$$

Assumption , the flow properties vary only with ϕ and are independent of r

$$\frac{\partial V_r}{\partial r} = \frac{\partial p}{\partial r} = \frac{\partial V_\phi}{\partial r} = 0$$

$$\rho V_r + \frac{d}{d\phi}(\rho V_\phi) = 0 \qquad V_\phi = \frac{dV_r}{d\phi} \qquad V_\phi\left(\frac{dV_\phi}{d\phi} + V_r\right) = -\frac{1}{\rho}\frac{dp}{d\phi}$$

$$\frac{dp}{d\phi} = \frac{dp}{d\rho}\frac{d\rho}{d\phi} = a^2\frac{d\rho}{d\phi}$$

$$\left(V_r + \frac{dV_\phi}{d\phi}\right)\left(1 - \frac{V_\phi^2}{a^2}\right) = 0$$

This is the governing equation for the Prandtl-Meyer flow

Let $V_r + \frac{dV_\phi}{d\phi} = 0 \longrightarrow \frac{dp}{d\phi} = 0$ From momentum eqn

The pressure is constant throughout the flow field and so there can be no expansion – ***Solution is Impossible*** since expansion field is considered

Possibility is $1 - \frac{V_\phi^2}{a^2} = 0 \qquad V_\phi = a$

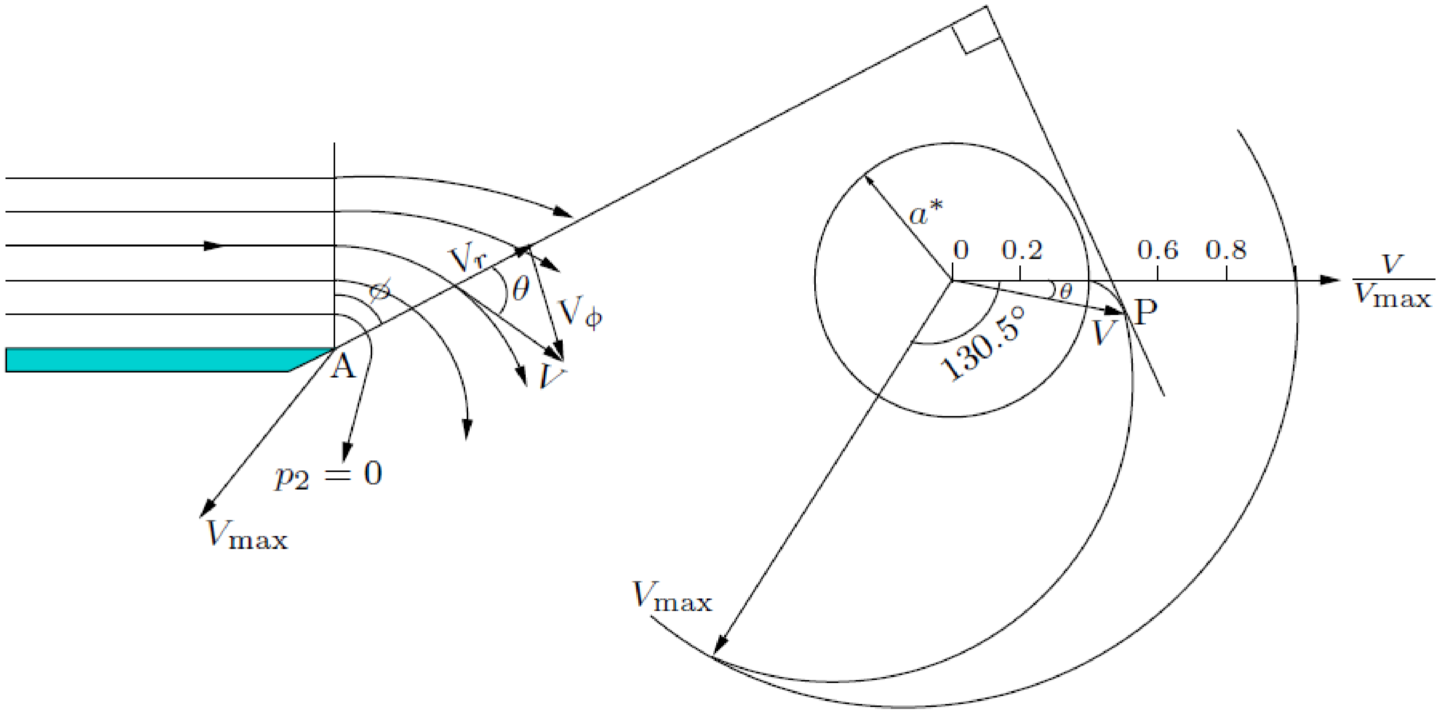
Inference
The velocity is constant along the radial lines and is equal to the local speed of sound. This means that the radial lines must correspond to *Mach lines*.

Mach angle is given by

$$\sin\mu = \frac{V_\phi}{V} = \frac{a}{V} = \frac{1}{M}$$

All rays in an expansion field are Mach lines and the entire flow is isentropic

Expansion fan Analysis in polar coordinate r and Φ



Velocity Components V_r and V_ϕ

compressible Bernoulli's equation

$$\frac{\gamma}{\gamma-1}\frac{p}{\rho}+\frac{V^2}{2}=\frac{\gamma}{\gamma-1}\frac{p_0}{\rho_0}=\frac{V_{\max}^2}{2}$$

$$V^2 = V_r^2 + V_\phi^2$$

$$V_\phi^2 = a^2 = \frac{\gamma p}{\rho}$$

$$V_r^2+\frac{\gamma+1}{\gamma-1}V_\phi^2=V_{\max}^2$$

Since the velocity and pressure are constants along the radial lines

$$V_r\frac{\partial V_r}{\partial r}+\frac{V_\phi}{r}\frac{\partial V_r}{\partial \phi}-\frac{V_\phi^2}{r}=-\frac{1}{\rho}\frac{\partial p}{\partial r} \quad \Longrightarrow \quad \frac{dV_r}{d\phi}=V_\phi$$

$$\frac{dV_r}{d\phi}=V_{\max}\sqrt{\frac{\gamma-1}{\gamma+1}}\sqrt{1-\left(\frac{V_r}{V_{\max}}\right)^2}$$

On integration by separation of variables

$$\phi\sqrt{\frac{\gamma-1}{\gamma+1}} = \arcsin\left(\frac{V_r}{V_{\text{max}}}\right) + \text{constant}$$

when $\phi = 0$, constant $= 0$

$$V_r = V_{\text{max}} \sin\left(\phi\sqrt{\frac{\gamma-1}{\gamma+1}}\right)$$

$$V_\phi = V_{\text{max}}\sqrt{\frac{\gamma-1}{\gamma+1}} \cos\left(\phi\sqrt{\frac{\gamma-1}{\gamma+1}}\right)$$

For $\phi = 0$

$$V_\phi = V_{\text{max}}\sqrt{\frac{\gamma-1}{\gamma+1}}$$

Substituting for V_{max} from Bernoulli's equation

$$V_\phi = \sqrt{\frac{2\gamma}{\gamma+1}\frac{p_0}{\rho_0}} = \sqrt{\frac{2}{\gamma+1}}\, a_0 = a^*$$

At the beginning of the fan the Φ-component of velocity corresponds to the speed of sound at sonic condition

Pressure at any point in the Prandtl-Meyer flow in terms of Φ

$$V_r^2 + V_\phi^2 = V_{max}^2\left[1-\left(\frac{p}{p_0}\right)^{\frac{\gamma-1}{\gamma}}\right]$$

Replacing V_r and V_ϕ

$$\left(V_{max}\sin\left(\phi\sqrt{\frac{\gamma-1}{\gamma+1}}\right)\right)^2 + \left(V_{max}\sqrt{\frac{\gamma-1}{\gamma+1}}\cos\left(\phi\sqrt{\frac{\gamma-1}{\gamma+1}}\right)\right)^2 = V_{max}^2\left[1-\left(\frac{p}{p_0}\right)^{\frac{\gamma-1}{\gamma}}\right]$$

$$\left(\frac{p}{p_0}\right)^{\frac{\gamma-1}{\gamma}} = \frac{1}{\gamma+1}\left[1+\cos\left(2\phi\sqrt{\frac{\gamma-1}{\gamma+1}}\right)\right]$$

The flow can be turned by an angle Φ_{max} where the pressure *p* will become zero

For this situation,

$$V_r = V_{max}, \quad V_\phi = 0$$

$$\phi_{max} = \frac{\pi}{2}\sqrt{\frac{\gamma+1}{\gamma-1}}$$

For a perfect gas, such as air with $\gamma = 1.4$, the maximum turning angle due to expansion is

$$\boxed{\phi_{max} = 220.5^\circ}$$

Thus, the limiting maximum flow turning angle associated with Prandtl-Meyer expansion is

$$\boxed{\theta = 130.5^\circ}$$

$$V^2 = V_r^2 + V_\phi^2 \qquad V_r^2 + V_\phi^2 = V_{max}^2 \left[1 - \left(\frac{p}{p_0}\right)^{\frac{\gamma-1}{\gamma}}\right]$$

The flow can be turned by an angle Φ_{max} where the pressure p will become zero

For this situation,

$$V_r = V_{max}, \quad V_\phi = 0$$

For a perfect gas, such as air with $\gamma = 1.4$, the maximum turning angle due to expansion is

$$\boxed{\phi_{max} = 220.5^\circ}$$

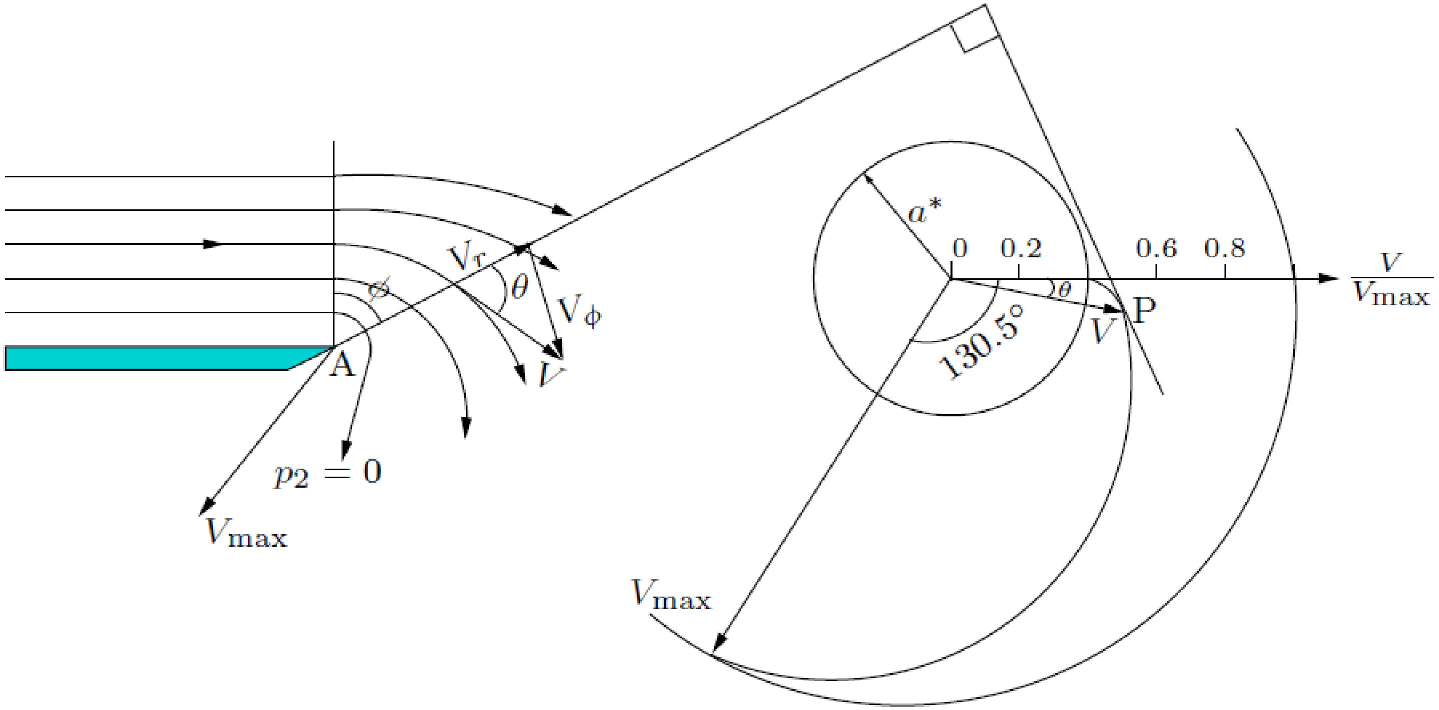
Thus, the limiting maximum flow turning angle associated with Prandtl-Meyer expansion is

$$\boxed{\theta = 130.5^\circ}$$

Inference

The beginning of the Prandtl- Meyer expansion is marked by Φ = 0, where the radial component of flow velocity is zero.

The expansion can go to the maximum extent where the static pressure of the flow goes to zero and the –component of the flow velocity also vanishes

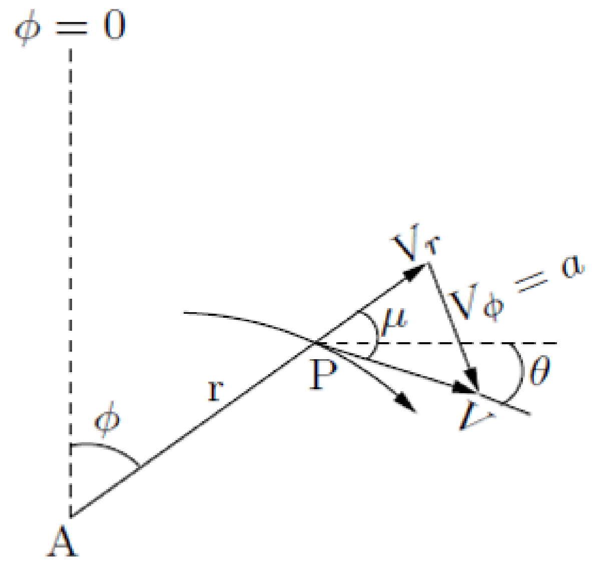


$$\theta = \phi + \mu - \pi/2$$

$$\phi = \pi/2 + \nu - \mu$$

$$\tan\mu = \frac{V_\phi}{V_r}$$

$$\cot\mu = \sqrt{M^2 - 1}$$

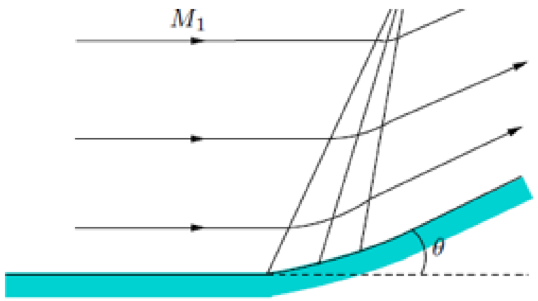


$$\nu = \sqrt{\frac{\gamma+1}{\gamma-1}} \arctan \sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} + \mu - \frac{\pi}{2}$$

$$\nu = \sqrt{\frac{\gamma+1}{\gamma-1}} \arctan \sqrt{\frac{\gamma-1}{\gamma+1}(M_1^2-1)} - \arctan \sqrt{(M_1^2-1)}$$

Physically, v is the flow inclination from $M_1 = 1$ (i.e. $\Phi = 0$) line

Isentropic compression analysis



Flow turning angle $\theta \leq 5°$ may be taken as the limit for considering the compression to be isentropic

In the proximity of the wall of a continuous compression, where the weak compressions waves generated retain their individual identity, the waves can be approximated as isentropic compression waves.

A Mach 2 air stream passes over a 10° isentropic compression corner. Find the downstream Mach number of the flow following the 10° turn.

Small values of θ , the compression through a single compression wave, the process can still be treated as isentropic

$$\nu_1 = 26.38^\circ$$

$$\nu_2 = \nu_1 - \theta = 16.38^\circ$$

$$M_2 = \boxed{1.652}$$

$$\nu_n = \nu_{n-1} + |\theta_n - \theta_{n-1}| \text{ (expansion)}$$

$$\nu_n = \nu_{n-1} - |\theta_n - \theta_{n-1}| \text{ (compression)}$$